

Note 1: The Apéry Generating Function ODE for $\zeta(3)$

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1 The Generating Function

Define the Apéry accelerated series:

$$F(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3 \binom{2n}{n}} x^n. \quad (1)$$

This series converges for $|x| \leq 4$ (the radius of convergence is $R = 4$).

Proposition 1. $F(1) = \frac{2}{5}\zeta(3)$.

Proposition 2. $F(x)$ satisfies the third-order linear ODE

$$x^2(4+x)F''' + x(10+3x)F'' + (2+x)F' = 1 \quad (2)$$

with initial conditions $F(0) = 0$, $F'(0) = \frac{1}{2}$, $F''(0) = -\frac{1}{24}$. All coefficients are integers. All initial values are rational.

Both propositions have been verified numerically to 13 decimal places.

2 Transformation to the t -Domain

To formulate an initial value problem on $[0, \infty)$, we use the substitution $x = 1 - e^{-t}$, so $x(0) = 0$, $\lim_{t \rightarrow \infty} x(t) = 1$, and $\dot{x} = 1 - x$.

Define state variables $y_0 = F$, $y_1 = F'$ (with respect to x), $y_2 = F''$. The chain rule gives:

$$\dot{y}_0 = y_1(1-x) \quad (3)$$

$$\dot{y}_1 = y_2(1-x) \quad (4)$$

$$\dot{y}_2 = F'''(x) \cdot (1-x) = \frac{(1-x)[1 - x(10+3x)y_2 - (2+x)y_1]}{x^2(4+x)} \quad (5)$$

Observation 1 (Boundedness). *Numerical integration confirms that all trajectories are bounded:*

$$y_0 \in [0, 0.481], \quad y_1 \in [0.463, 0.500], \quad y_2 \in [-0.042, -0.033], \quad x \in [0, 1].$$

The convergence rate is $\alpha \approx 1.0001$, i.e., first-floor convergence.

Observation 2 (Rational initial conditions). *The initial conditions at $t = 0$ (i.e., $x = 0$) are:*

$$y_0(0) = 0, \quad y_1(0) = \frac{1}{2}, \quad y_2(0) = -\frac{1}{24}, \quad x(0) = 0.$$

All rational. No transcendental numbers appear in the initial conditions.

3 The Obstacle: Regular Singular Point

The equation for $\dot{y}_2$ contains the factor $1/[x^2(4+x)]$, which diverges at $x = 0$. This is the **only** obstruction to the system being a polynomial PIVP.

The singularity at $x = 0$ is a *regular singular point* (Fuchsian) of the ODE (??). The solution $F(x)$ is analytic at $x = 0$ — the singularity is an artifact of the coefficient structure, not a true blow-up.

Concretely, at $x = 0$:

- Numerator of F''' : $1 - 0 \cdot (\dots) \cdot y_2 - 2 \cdot y_1 = 1 - 2 \cdot \frac{1}{2} = 0$.
- Denominator: $x^2(4+x) = 0$.

The ratio $0/0$ has a finite limit: $F'''(0) = 6a_3 = 6 \cdot \frac{1}{540} = \frac{1}{90}$.

So the system is *numerically* well-behaved (starting from a nearby point works perfectly), but *formally* it is not a polynomial ODE system.

4 The Core Question

Open Problem 1. Can the regular singular point at $x = 0$ be regularized — via variable substitution, blow-up, desingularization, or any other algebraic technique — to obtain a polynomial PIVP with rational initial conditions that computes $\zeta(3)$?

If yes, then $\zeta(3) \in R_{\text{RTCRN}}$.

4.1 Why This Matters

Compared to the integral-based PIVP (which computes $\zeta(3)$ but requires transcendental ICs involving e and $1/(e-1)$), this generating-function ODE has:

1. Rational initial conditions (no transcendental numbers).
2. Integer coefficients in the ODE.
3. First-floor convergence.
4. Bounded trajectories.

The *only* missing ingredient is polynomiality of the right-hand side.

4.2 Possible Approaches

1. **Variable substitution:** Define new variables (e.g., $G = F/x$, $H = x^k F^{(j)}$) to absorb the x^2 factor. Challenge: substitutions that remove x^2 from the denominator tend to introduce $1/x$ elsewhere.
2. **Desingularization (blow-up):** Standard technique from algebraic geometry for resolving singular points of ODEs. Replace the singular point with a higher-dimensional smooth manifold.
3. **Regularization by time reparametrization:** Instead of $x = 1 - e^{-t}$, use $dt/ds = x^2(4+x)$ (i.e., slow down time near the singularity). This makes the RHS polynomial, but the new time variable s may not reach $x = 1$ in finite s -time, or the system may become unbounded.
4. **Find a different ODE:** Perhaps there exists another ODE (possibly higher order or with more variables) for a related function that has no singular point and still computes $\zeta(3)$.

5 Summary

The generating function of the Apéry accelerated series provides a bridge between the discrete series $\zeta(3) = \sum 1/k^3$ and a continuous ODE. The resulting ODE has all the properties needed for a bounded GPAC computation of $\zeta(3)$ *except* polynomiality at a single regular singular point. Resolving this singularity would settle the question of whether $\zeta(3) \in R_{\text{RTCRN}}$.